

Arunabha Majumder

Mechanical Engineer — Robotic Actuators, Joint Kinematics, Prototyping & Test Rigs

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SUMMARY

Mechanical engineer who designs robotic actuators, joints and kinematic mechanisms and builds them into working hardware. Designed a robotic shoulder actuation system that cut energy use by more than 25% (Best Paper Award, 2026), modelled parallel-robot kinematics to below 10% force-estimation error, and builds own prototypes and test rigs. Works fast: first prototype early, then iterate.

KEY RESULTS

- **>25%** energy reduction from a new actuator and transmission concept
- **<10%** RMS error from Jacobian-based force estimation on a 5-bar parallel robot
- **CAD** → **prototype** → **test rig**: robotic joints and actuators designed, built and tested in-house

EXPERIENCE

Research Engineer – Robotic Actuators & Mechanical Design (PhD Project: VIEXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026) by designing the actuator and transmission concept for a robotic shoulder joint: a parallel spring working with a 6 Nm motor, sized so torque and load requirements were met across the assistance cycle.
- Took a robotic joint from CAD to working hardware, iterating quickly from a first prototype, by designing mechanical components and assemblies in SolidWorks and Fusion 360 and making parts through additive manufacturing and machining.
- Improved strength-to-weight before manufacture by running stress analysis (FEA) for stiffness, strength and weight optimisation of load-carrying parts.
- Achieved below 10% RMS error in end-effector force estimation on a 5-bar planar parallel robot, verified against a force sensor, by modelling its kinematics (Jacobian) and variable-stiffness actuators.
- Integrated electronics into the mechanical assembly (motor, encoder, force sensors, DAQ and embedded control), then built test rigs and tracked down problems on the bench through repeated test campaigns, including tests under 2 load conditions.
- Kept several projects moving at once (two robots, testing, a 12-participant data collection, and mentoring Bachelor's students in 2023 – 2025) with full design documentation: 3D models, drawings and part specifications.

MTech Researcher – Pneumatic Actuators (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Characterised actuator load–displacement behaviour by building pneumatic artificial muscles in-house from raw materials and an actuator test rig with valves, pressure sensors and DAQ.
- Reduced position-tracking error to 0.3–0.78% with a neural-network-based controller, outperforming classical PID under varying load.

THINGS I BUILT

- **Exoskeleton joint (2023–present)**: Hybrid-actuated robotic shoulder joint: actuator concept, CAD, FEA, fabrication, electronics, test rigs.
- **5-bar parallel robot (PhD)**: Kinematic model and force estimation from variable-stiffness actuators, tested in the gait lab.
- **Rehabilitation exoskeleton (2018–2019)**: Concept, CAD, fabrication and user testing; Best Major Project Award and TATA Technologies Innovation Grant.

- **Go-Kart (2017):** Designed, fabricated and assembled from the ground up.

SKILLS

- **Robotics mechanics:** Actuator selection and sizing (motors, encoders), joint kinematics, transmissions, parallel mechanisms
- **CAD:** SolidWorks (CSWA), Autodesk Fusion 360, assemblies, drawings, part specifications
- **Analysis & manufacturing:** FEA stress analysis, weight optimisation, additive manufacturing, machined parts, DFA
- **Prototyping & testing:** Test rigs, load testing, bench troubleshooting, sensor and DAQ integration, embedded control
- **Software:** MATLAB/Simulink, Python, Arduino, LabVIEW (basic)
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Two peer-reviewed publications: ISRM 2026 (Springer) and IEEE CONECCT 2022

REFERENCES

Available upon request.